

Section 1 – Core Machine Learning Concepts (Q1 – Q10)

1. What is the Bias–Variance Trade-off?

- Bias means your model is too simple to capture the underlying trend. For example, predicting house prices using only square footage while ignoring location or amenities gives consistently wrong estimates — that’s high bias.
 - Variance means your model reacts too strongly to small fluctuations in the training data. A very deep decision tree might perfectly fit the training set but fail badly on new listings because it has memorized noise.
 - The trade-off is about finding the balance between the two — a model just complex enough to learn real patterns but not the noise. Regularization, cross-validation, and ensemble methods are common ways to manage this balance.
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2. What is the Curse of Dimensionality?

- As you add more features, the data space expands rapidly and points become increasingly far apart. Distance-based algorithms like KNN or clustering then struggle because “near” and “far” start to look the same.
 - It also means you need exponentially more data to cover all possible combinations of feature values, making model learning inefficient or unstable.
 - In practice, we reduce dimensionality using methods such as PCA, autoencoders, or domain-driven feature selection — for instance, in a credit model, combining multiple income sources into one “total income” variable keeps information but lowers dimensionality.
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3. What are the Limitations of the KNN Algorithm?

- KNN makes predictions by comparing a new data point to every example in the training set, so it’s slow at prediction time, especially when the dataset is large.
 - It requires scaling because features measured on larger numeric ranges dominate the distance measure — e.g., income in lakhs can overshadow age in years.
 - It performs poorly on high-dimensional data or when there’s noise and irrelevant features. In real life, KNN works well for small, clean datasets such as image pattern matching or recommendation prototypes, but not for millions of records.
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4. In a breast-cancer detection problem (1000 images, 80:20 split), how do you ensure the model learns well?

- Always maintain class balance with a stratified train-test split, so both malignant and benign cases are proportionally represented.
 - Use data augmentation — flips, rotations, or brightness adjustments — to expand the training set without new data collection.
 - Employ transfer learning by fine-tuning a pre-trained CNN like ResNet or MobileNet rather than training from scratch.
 - Evaluate through cross-validation and use a validation set to monitor overfitting. Fit all preprocessing (scaling or normalization) only on training data to prevent leakage.
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5. Why do we not scale the test dataset independently from the training dataset?

- Scaling must follow the same statistical parameters (mean, standard deviation, min, max) computed from the training data to ensure a consistent feature range.
- If you scale the test data independently, you use information from unseen data, causing data leakage and unrealistic results.

- The correct approach: fit your scaler on training data, then reuse it to transform both training and test sets — just as you would in real deployment where new data follows existing scaling rules.

6. Which scaling technique is more sensitive to outliers?

- StandardScaler and Min-Max Scaler are sensitive because both rely on the mean and standard deviation or the extreme values; a single outlier can shift the entire scale.
- RobustScaler uses the median and inter-quartile range, making it ideal when your data contains occasional extreme points.
- In financial datasets with a few very high-value customers, RobustScaler or a log transformation prevents distortion while keeping meaningful relationships intact.

7. Why is scaling a necessary pre-processing step?

- Many models, especially those that use distance or gradients (KNN, SVM, Logistic Regression, Neural Networks), assume all features are on comparable scales.
- Without scaling, features measured in different units (e.g., “income in lakhs” vs “age in years”) can mislead the model about which variable matters more.
- It also helps gradient-descent optimization converge faster and ensures regularization treats all features fairly. In practice, consistent scaling is part of every production ML pipeline.

8. What is the Least Squares Algorithm and which popular technique uses it?

- The Least Squares method finds the line (or hyperplane) that minimizes the total squared difference between predicted and actual values.
- It forms the basis of Linear Regression, which tries to best explain the relationship between inputs and target through the smallest possible residuals.
- For example, in predicting house prices from square footage, the algorithm positions the regression line so overall prediction errors are minimized in the squared sense — giving more weight to larger mistakes.

9. What is a Residual and why is it important?

- A residual is simply the gap between what the model predicted and what actually happened — $\text{actual} - \text{predicted}$.
- Studying residuals helps you see where your model fails: a random scatter of residuals indicates a good fit, but clear patterns show that important relationships or nonlinearities are missing.
- Residual plots are a simple yet powerful diagnostic; for example, if residuals grow larger with increasing house size, it suggests the model underestimates for large houses and needs a non-linear term.

10. How do you interpret the influence of a predictor variable on the label?

- In linear or logistic regression, the sign of a coefficient tells you the direction of impact (positive or negative) and the magnitude shows the strength of influence after scaling.
- For logistic regression, exponentiating the coefficient gives an odds ratio — easy to explain to business users (“each additional year of customer tenure reduces churn odds by 20 %”).
- For complex models like Random Forest or XGBoost, we rely on feature importance, SHAP values, or Partial Dependence Plots to understand which variables drive outcomes.

- Real-world use case: marketing teams interpret these insights to focus campaigns on the most influential drivers of customer conversion.
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Section 2 – Model Evaluation & Data Issues (Q11–Q15)

11. How do you remove multicollinearity?

- Multicollinearity happens when two or more input variables carry overlapping information. For example, “monthly income” and “annual income” are nearly identical and confuse the model about which feature to trust.
 - You can detect it by checking correlation matrices or using Variance Inflation Factor (VIF) — a high VIF (>5 or 10) signals redundancy.
 - To fix it, you can drop one of the correlated features, combine them into one (like total expenditure = spend on A + spend on B), or use a regularized algorithm (Ridge regression) that penalizes redundant weights.
 - In finance or credit-risk models, multicollinearity often arises among “loan amount,” “loan-to-income,” and “debt-to-income” — simplifying these helps stabilize predictions and interpretation.
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12. Why do we measure training accuracy and how do we detect overfitting?

- Training accuracy shows whether the model learned the patterns in its own dataset. If even training accuracy is low, the model is too simple (underfitting).
 - Overfitting occurs when the model memorizes training details instead of generalizing — it performs brilliantly on known data but poorly on new cases.
 - You can spot overfitting by comparing training vs validation/test accuracy or by visualizing learning curves. A large performance gap means the model learned noise.
 - Common fixes include regularization, early stopping, simpler models, or data augmentation. For example, a churn-prediction model giving 99 % train accuracy but 78 % test accuracy is overfitting and needs stronger regularization or fewer features.
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13. What are the limitations of linear regression models?

- Linear regression assumes a straight-line relationship between inputs and output. Many business processes — like advertising spend vs. sales — have diminishing returns or curved effects, which linear models can’t capture.
 - It is sensitive to outliers; one unusually high or low observation can shift the regression line significantly.
 - It also struggles when predictors are highly correlated (multicollinearity) or when error variance is not constant across data ranges.
 - Because of these limitations, linear regression is often used as a baseline for interpretability, but tree-based or ensemble models usually perform better for complex, real-world data.
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14. Why is accuracy not a good metric for imbalanced datasets or multi-class problems?

- Accuracy can look deceptively high when one class dominates. In a fraud-detection dataset with 99 % normal transactions, predicting “no fraud” every time yields 99 % accuracy but zero value.

- It hides how well the model handles minority classes, which are usually the most important in business.
 - Metrics like Precision, Recall, F1-Score, or AUC better represent performance on rare but critical cases.
 - For example, in medical screening, missing a few positive cases (low recall) can be far worse than a few false alarms (lower precision). Always pick a metric that matches the business objective.
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15. What probability-based linear models do you know, and why are they probability-based?

- Models such as Logistic Regression, Naïve Bayes, and Linear Discriminant Analysis (LDA) estimate the probability of each outcome instead of giving a hard class label.
 - These probabilities allow flexible decision-making — for example, a bank can flag only applications with a default risk above 0.7 for manual review.
 - Logistic Regression converts linear combinations of inputs into a 0–1 probability using a sigmoid function; Naïve Bayes multiplies the probabilities of each feature given a class.
 - In practice, probability-based models are widely used in churn prediction, lead scoring, and marketing-response models because they let teams prioritize by likelihood, not just classification.
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Section 3 – Regularization & Generalized Linear Models (Q16–Q24)

16. What is Regularization? How do L1 and L2 regularization differ?

- Regularization helps prevent overfitting by penalizing large model coefficients, keeping the model simpler and more generalizable.
 - L1 Regularization (Lasso) adds an absolute value penalty to the loss function, forcing some coefficients to exactly zero — this acts as automatic feature selection.
 - L2 Regularization (Ridge) adds a squared penalty that shrinks all coefficients toward zero smoothly but rarely makes them exactly zero — better for models with correlated predictors.
 - In business use, Ridge stabilizes financial risk models with correlated income indicators, while Lasso is great for text or marketing data with hundreds of sparse features.
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17. How does Ridge prevent overfitting differently from Lasso?

- Ridge reduces overfitting by distributing weight across many correlated features instead of removing them entirely.
 - Lasso, on the other hand, performs feature elimination, dropping weaker predictors to make a sparse and interpretable model.
 - Ridge is ideal when every variable carries some information — like predicting energy demand using many interrelated weather factors.
 - Lasso is preferred when you have a large number of weak predictors and need a compact model — like selecting key terms in a sentiment analysis model.
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18. What is ElasticNet, and when would you prefer it over Lasso or Ridge?

- ElasticNet blends the benefits of both L1 (sparsity) and L2 (stability) regularization. It's controlled by two parameters that balance how much of each penalty to apply.
 - It's particularly useful when you have many correlated variables because Lasso may pick one arbitrarily and ignore others, while ElasticNet retains groups of related features.
 - Real-world example: in customer behavior modeling, multiple spending categories may correlate — ElasticNet keeps all related predictors but controls their combined effect.
 - It's often the safest "default" when you're unsure whether L1 or L2 will perform better.
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19. Why does Logistic Regression use a sigmoid instead of a linear output?

- The sigmoid function maps any numeric input into a range between 0 and 1, making it ideal for expressing probabilities.
 - Linear models can output values far outside this range, which don't make sense for a probability.
 - The sigmoid also provides a smooth gradient, making optimization stable and allowing probability interpretation.
 - Example: predicting customer churn probability — a sigmoid output of 0.82 clearly means an 82 % chance of churn, which is easy to explain to business stakeholders.
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20. What is the difference between Log Loss and Cross-Entropy Loss?

- For binary classification, log loss and binary cross-entropy are essentially the same; both measure how well predicted probabilities match the true labels.
 - They heavily penalize confident but wrong predictions — predicting 0.99 for a negative class hurts far more than predicting 0.6.
 - Cross-entropy generalizes this idea to multi-class problems, summing the loss across all classes.
 - In marketing-response prediction, minimizing log loss ensures your probability estimates are well-calibrated for campaign targeting.
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21. What evaluation metric do you prefer for imbalanced classification, and why?

- In imbalanced datasets, accuracy can be misleading because the model can ignore minority cases and still appear good.
 - If the goal is to detect rare events (fraud, churn, disease), Recall or F1-Score is more meaningful.
 - If false alarms are expensive (e.g., manual reviews in banking), focus on Precision.
 - PR-AUC (Precision-Recall AUC) gives a better overall sense of performance on rare positives than ROC-AUC.
 - For a churn model, F1 or PR-AUC is typically the most insightful metric because it rewards models that find churners without too many false positives.
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22. What are Precision, Recall, and F1-Score? Explain with an example.

- Precision measures how many predicted positives are actually correct — high precision means fewer false alarms.
- Recall measures how many of the actual positives your model successfully identified — high recall means you're catching most of the important cases.
- F1-Score is the harmonic mean of Precision and Recall — it gives a balanced view when both false positives and false negatives matter.

- Example: in a fraud detection system, you might have 80 % precision and 50 % recall — good at being correct but missing many frauds. Improving recall (catch more frauds) may slightly reduce precision but increase overall business value.

23. What is the ROC Curve, and what does AUC tell you?

- The ROC Curve plots the True Positive Rate (Recall) against the False Positive Rate at various thresholds, showing how the model's sensitivity changes.
- The AUC (Area Under Curve) summarizes this performance — an AUC of 1.0 means perfect discrimination, while 0.5 means random guessing.
- It helps compare models independent of a specific threshold.
- For example, when choosing between two churn models, the one with higher AUC consistently ranks churners above non-churners across all possible thresholds.

24. What is the difference between ROC-AUC and PR-AUC, and which one should you use for churn modeling?

- ROC-AUC considers both positive and negative classes, but it can look overly optimistic when the positive class is rare.
- PR-AUC (Precision–Recall AUC) focuses only on the positive class and is more informative for imbalanced problems.
- In churn modeling, where churners are a small minority, PR-AUC gives a more realistic picture of how well the model finds true churners without too many false alarms.
- A practical rule: use ROC-AUC for balanced datasets, PR-AUC when positive cases are rare but important.

Section 4 – Tree-Based Models & Ensemble Learning (Q25–Q32)

25. How does a Decision Tree decide the best split?

- A decision tree divides data based on feature thresholds that create the purest possible child nodes — meaning the least mixed labels.
- It evaluates all possible splits using metrics like Gini Impurity (used in CART) or Entropy/Information Gain (used in ID3 or C4.5).
- The goal is to choose the split that best separates the classes or reduces prediction error.
- Example: In a telecom churn model, the first split might be on “Contract Type” because customers with month-to-month contracts have a distinctly higher churn rate, making that split most informative.

26. What is pruning, and why do trees tend to overfit?

- Decision trees can grow very deep, fitting even random fluctuations in the training data — this is classic overfitting.
- Pruning limits tree growth by cutting back branches that add little predictive value.
- There are two approaches: pre-pruning (set limits like max depth or min samples per leaf) and post-pruning (build a full tree, then remove weak splits based on validation performance).

- Example: A customer segmentation tree that keeps splitting until each group has one customer will perform perfectly on training but fail on new data — pruning keeps it practical and generalizable.

27. What is Bootstrap Aggregation (Bagging)?

- Bagging is an ensemble technique that trains multiple models on different random subsets (with replacement) of the training data.
- Each model votes (for classification) or averages (for regression) to form a final prediction.
- This reduces variance, as errors from individual models tend to cancel out when aggregated.
- For example, instead of relying on one decision tree, bagging builds hundreds on slightly different samples — forming the foundation for Random Forests.

28. How does Random Forest reduce variance compared to a single tree?

- A Random Forest trains many decision trees on bootstrapped samples and adds another layer of randomness — each tree can only use a random subset of features for splitting.
- This decorrelates trees, so their prediction errors are not aligned, reducing overall variance and making the ensemble much more stable.
- The final prediction is a majority vote (classification) or average (regression) of all trees.
- In real-world terms, it's like getting opinions from 100 independent experts rather than trusting one who might be biased.

29. What is Out-of-Bag (OOB) error, and why is it useful?

- In bagging, roughly one-third of the data isn't used in training each tree — those unused samples are Out-of-Bag (OOB) for that tree.
- We can test the tree's performance on its OOB data to get a built-in, unbiased estimate of model error without needing a separate validation set.
- OOB error is especially handy when data is limited, saving time and data splits.
- In practice, you can check Random Forest's OOB score to quickly estimate its generalization performance.

30. Compare Random Forest vs. Gradient Boosting — when would you use which?

- Random Forest builds trees independently in parallel, while Gradient Boosting builds trees sequentially, each one correcting the mistakes of the previous.
- Random Forest is robust, simple to tune, and rarely overfits — perfect for baseline models and when you need quick, reliable results.
- Gradient Boosting (e.g., XGBoost, LightGBM) achieves higher accuracy but requires careful tuning of learning rate, number of estimators, and depth.
- In business, you might start with Random Forest to get quick insights, then move to Gradient Boosting for production where every extra percentage of accuracy matters.

31. What is Gradient Boosting, and why does it fit trees sequentially?

- Gradient Boosting builds trees step by step, where each new tree focuses on the residual errors (the difference between prediction and actual values) left by the previous trees.
 - This sequential process allows the model to learn complex patterns gradually, improving performance while keeping each individual tree weak (to avoid overfitting).
 - Think of it as teaching a model to “correct its own mistakes” iteratively.
 - Example: in a loan default model, the first tree might capture income trends, the second focuses on repayment history, and later ones fine-tune subtle factors like credit utilization.
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32. How does XGBoost differ from the traditional GradientBoostingClassifier?

- XGBoost (Extreme Gradient Boosting) enhances traditional boosting with advanced regularization (L1/L2), efficient parallelization, and smart handling of missing values.
 - It uses a more optimized gradient computation and shrinkage (learning rate) to prevent overfitting while training much faster.
 - It also includes early stopping, allowing the model to halt training when validation performance stops improving.
 - That’s why XGBoost became a go-to choice for Kaggle competitions and enterprise ML — it delivers speed, scalability, and accuracy on large tabular datasets.
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Section 5 – Clustering & Dimensionality Reduction (Q33–Q39)

33. What are feature importances, and how reliable are impurity-based ones?

- Feature importance tells us how much each input variable contributes to model performance. In tree-based models, it’s usually computed from how much each feature reduces impurity (like Gini or entropy) across all splits.
 - However, impurity-based importance can be biased toward features with many unique values or continuous ranges — for example, “Customer ID” might falsely appear important simply because it splits data finely.
 - A better way is permutation importance, which measures how much the model’s performance drops when a feature’s values are randomly shuffled.
 - In practice, impurity importance is fine for early exploration, but for serious interpretation, always cross-check with SHAP values or permutation methods.
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34. Explain the concept of Support Vectors in SVM.

- In Support Vector Machines (SVM), support vectors are the data points closest to the decision boundary — they are the “critical” examples that define where the separating line (or hyperplane) should lie.
 - Only these points matter; the others are irrelevant to defining the margin. That’s why SVMs are called sparse models — they depend only on a subset of training data.
 - Real-world example: In spam detection, the emails that are borderline between spam and non-spam (support vectors) define the classification boundary far more than obviously spammy ones.
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35. What is the kernel trick, and why do we need kernel functions?

- Some datasets cannot be separated by a straight line in their original feature space — this is where the kernel trick comes in.
 - The kernel function allows SVM (or similar models) to implicitly project data into a higher-dimensional space where it becomes linearly separable, without explicitly computing that transformation.
 - Common kernels include linear, polynomial, and RBF (Radial Basis Function).
 - Example: Classifying concentric circles — impossible with a straight line in 2D, but an RBF kernel maps them into a higher dimension where a simple boundary works perfectly.
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36. Why is SVM not ideal for large datasets?

- SVM training involves calculating a kernel matrix of pairwise similarities between all data points — that's an $O(n^2)$ operation, which becomes impractical as n (the number of samples) grows into the millions.
 - It also requires significant memory and time for optimization because SVM must find the optimal hyperplane by solving a large quadratic programming problem.
 - For massive datasets, Linear SVMs, Logistic Regression, or tree-based ensembles (like XGBoost) are usually faster alternatives with comparable performance.
 - In real-world projects like fraud detection or ad-click prediction, SVMs are rarely used due to scalability constraints.
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37. What is clustering, and how is it different from classification?

- Clustering is an unsupervised learning technique that groups similar data points based on feature similarity without any predefined labels.
 - Classification, in contrast, is supervised — the model learns from labeled examples to predict known categories.
 - Clustering is used for pattern discovery — for example, segmenting retail customers into high-, medium-, and low-value groups without knowing those groups beforehand.
 - It's an exploratory tool that helps generate business insights, while classification is about making precise predictions.
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38. Explain how K-Means works and why it requires feature scaling.

- K-Means begins by randomly assigning k centroids, then iteratively assigns each data point to the nearest centroid and updates those centroids based on the mean of assigned points.
 - The algorithm repeats this until centroids stabilize and cluster memberships no longer change significantly.
 - It relies on Euclidean distance, which is scale-sensitive — features with large numeric ranges dominate the clustering.
 - That's why feature scaling (e.g., StandardScaler) is mandatory. For instance, in customer segmentation, unscaled income (in lakhs) can overshadow age or frequency if not standardized.
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39. Why does K-Means struggle with non-spherical clusters?

- K-Means assumes clusters are spherical and evenly sized, with clear center points — it minimizes within-cluster variance, not shape complexity.
- If clusters are elongated, overlapping, or of varying density (like moon-shaped or spiral clusters), K-Means assigns points incorrectly because the Euclidean distance doesn't reflect true similarity.

- Alternative algorithms like DBSCAN, Gaussian Mixture Models, or Spectral Clustering handle such shapes better.
 - In real-world anomaly detection or image segmentation, DBSCAN often works better because it can find irregular or nested cluster shapes automatically.
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Section 6 – Metrics & Model Interpretation (Q40–Q43)

40. What is the Silhouette Score?

- The Silhouette Score measures how well each data point fits into its assigned cluster compared to other clusters. It combines both cohesion (how close a point is to its own cluster) and separation (how far it is from the nearest different cluster).
 - The score ranges from -1 to $+1$, where values near $+1$ mean the sample is well-clustered, 0 means it lies between clusters, and negative values suggest it's in the wrong cluster.
 - It's a great metric to decide the optimal number of clusters (k) in algorithms like K-Means when no labels exist.
 - Real-world example: A marketing analyst can test different k values using the Silhouette Score to find the most meaningful customer segmentation (e.g., "3 distinct customer types" gives the best average score).
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41. What is PCA (Principal Component Analysis), and why is it used in ML pipelines?

- PCA is a dimensionality reduction technique that transforms correlated features into a smaller set of uncorrelated components, called principal components.
 - It works by capturing directions of maximum variance in the data — so it keeps the most important patterns while discarding redundant or noisy information.
 - PCA helps in reducing computation time, removing multicollinearity, and improving model efficiency, especially in high-dimensional datasets.
 - Example: In image compression or face recognition, PCA reduces thousands of pixel features into a few key "eigenfaces" while retaining most of the visual identity.
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42. Does PCA guarantee improved model performance?

- Not always — PCA optimizes for variance, not necessarily for predictive accuracy.
 - It's beneficial when you have too many correlated features, which can confuse the model or inflate training time.
 - However, in some cases, PCA can reduce interpretability and may even drop low-variance features that carry important signal for prediction.
 - In business terms: use PCA when your dataset is wide and noisy (e.g., 500+ marketing campaign features) but test both versions — with and without PCA — to confirm if performance truly improves.
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43. What is a Confusion Matrix, and how do you interpret each cell?

- A Confusion Matrix is a 2×2 (or larger) table that summarizes a classifier's performance by comparing actual and predicted labels.
- It includes:
- True Positives (TP): Correctly predicted positive cases (e.g., churners correctly identified).

- False Positives (FP): Incorrectly predicted positives (e.g., loyal customers wrongly flagged as churners).
- True Negatives (TN): Correctly predicted negatives (e.g., loyal customers correctly predicted to stay).
- False Negatives (FN): Missed positive cases (e.g., churners missed by the model).
- From these, we derive key metrics like Precision, Recall, F1-Score, and Specificity.
- Real-world example: In a medical test, False Negatives are more dangerous (missed diagnoses), so we prioritize high Recall even if Precision drops slightly.